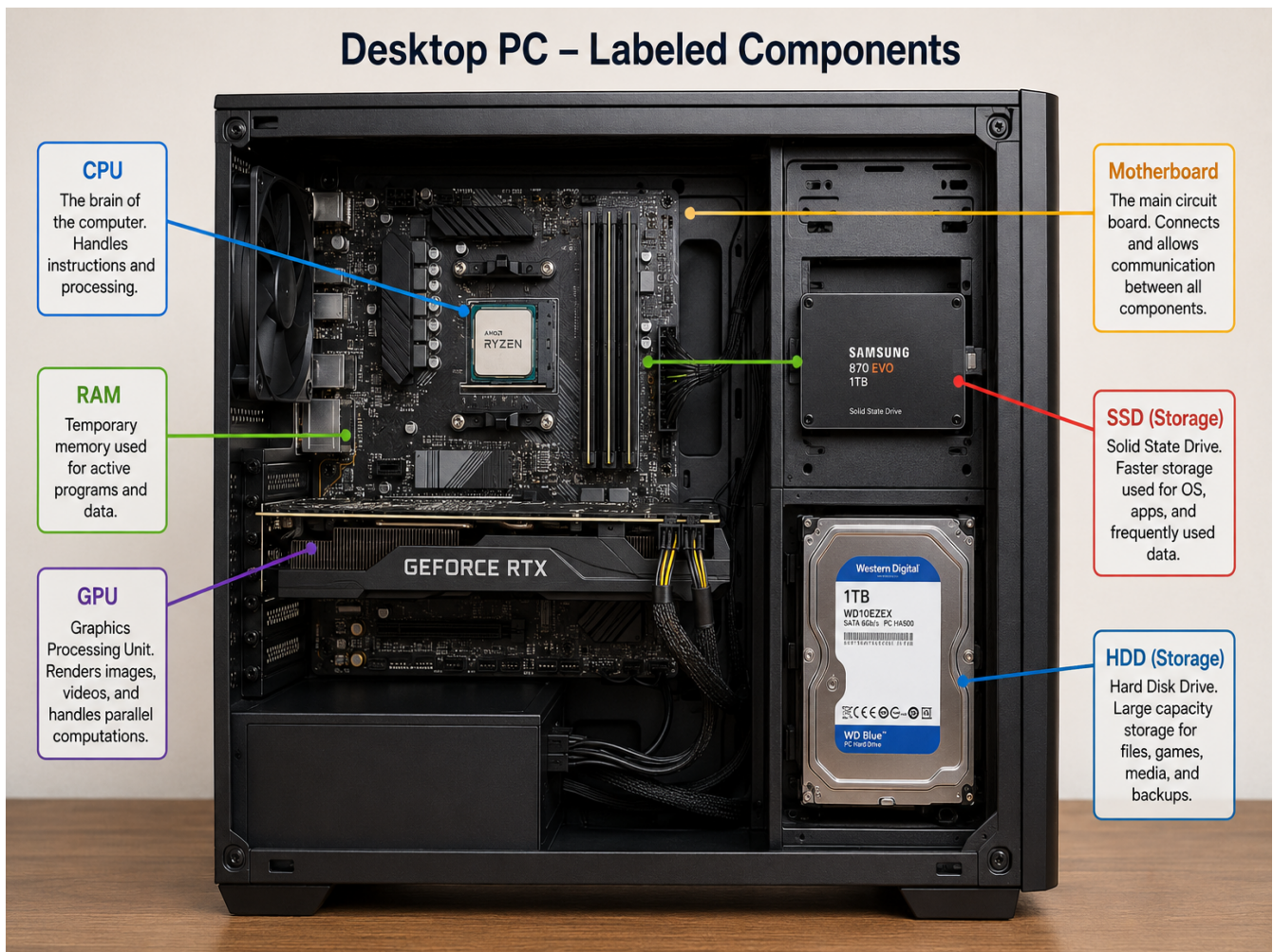


PC Hardware Assignment

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

SSD (Solid State Drive)	HDD (Hard Disk Drive)
Faster Speed	Slower Speed
More Durable	Less Durable
More Shock Resistant	Less Shock Resistant

No Moving Parts	Has Moving Parts
Common in Laptops & Gaming PCs	Common in Desktop PCs & Storage Systems
Higher Cost	Lower Cost

3. Write a short explanation (2–3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit): The CPU is called the brain of the computer. It processes all the instructions and controls game activities such as player movement, enemy actions, and game logic. A faster CPU helps the game run smoothly.

RAM (Random Access Memory): RAM stores temporary data while the game is running. It allows the game to load maps, characters, and other resources quickly. More RAM helps reduce lag and improves multitasking.

GPU (Graphics Processing Unit): The GPU is responsible for creating and displaying graphics. It renders game visuals, effects, and animations. A powerful GPU provides better graphics quality and higher frame rates.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

1. GPU (Graphics Card)

The GPU is one of the most important components for gaming. It is responsible for rendering game graphics, animations, shadows, and visual effects. Upgrading the GPU will improve graphics quality, increase FPS (Frames Per Second), and provide a smoother gaming experience while playing FIFA.

2. RAM (Random Access Memory)

RAM stores the data that the game needs while it is running. If the system has insufficient RAM, the game may lag or take longer to load. Upgrading RAM allows the game to run more smoothly, reduces stuttering, and improves overall system performance.

3. SSD (Solid State Drive)

An SSD is much faster than a traditional HDD. Installing FIFA on an SSD will significantly reduce game loading times, speed up system booting, and improve the overall responsiveness of the PC.

4. CPU (Central Processing Unit)

The CPU handles game calculations, player movements, AI behavior, and background processes. A faster and more powerful CPU ensures that the game runs efficiently without

performance bottlenecks, especially during intense gameplay situations.

5. Power Supply Unit (PSU)

When upgrading components such as the GPU and CPU, the system may require more power. A good-quality PSU provides stable electricity to all components and helps prevent crashes, overheating, and hardware damage.